

Raimundo Oscar de Sousa Jr.

Geophysicist

CONTACT

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Location
Curitiba, PR, Brazil

GitHub
github.com/raimundo1994

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linkedin.com/in/oscarfilho

LANGUAGES

Portuguese — Native
English — Advanced

PROGRAMMING

Python Julia
MATLAB SQL
L^AT_EX C++

DATA VISUALIZATION

Plotly PyVista
Matplotlib
Bokeh Seaborn
Power BI

GEOPROCESSING

QGIS ArcGIS
Leapfrog
Oasis Montaj
Google Earth Engine

OFFICE / DESIGN

Excel Word
PowerPoint
Canva Figma

EQUIPMENT

Electrical: SuperSting, Polares-32, Tectrol
Electromagnetic: GPR GSSI and LEICA DS200
Magnetic: GEM-19, G-857
Seismic: Geode
Others: Scintrex, Total Station, RTK

PROFESSIONAL SUMMARY

Geophysicist with over 5 years of experience in acquisition, processing and interpretation of geophysical data (electrical, electromagnetic, magnetic and seismic methods). Master's degree from the National Observatory (Brazil), with research focused on potential methods. Solid background in mineral exploration, environmental studies and civil construction, combined with skills in programming, data visualization and analysis.

PROFESSIONAL EXPERIENCE

Geophysicist II 2026– present
GEOFSPAN - Barueri, SP, Brazil

- Responsible for the management and operation of geophysical mapping projects using the electromagnetic method GPR (Ground Penetrating Radar) integrated with RTK (Real Time Kinematic) for high-precision georeferencing. Planning and coordination of field campaigns for GPR data acquisition, defining survey parameters (antenna frequency, line spacing, investigation depth) according to project objectives.

Geophysicist II 2023–2026
Signal Geophysics LTD — Curitiba, PR, Brazil

- Responsible for the complete technical execution of multi-method geophysical surveys, including acquisition, processing and integrated interpretation of data from: VES (Vertical Electrical Sounding) and ERT, GPR, Magnetometry, Seismic Refraction and Reflection. Technical management and leadership of multidisciplinary projects focused on mineral exploration, environmental studies, geotechnics, and risk analysis. Coordination of field and office teams, planning of scope, budget, schedule and logistics for concurrent projects. Preparation of technical and commercial proposals, as well as executive reports and conclusive opinions.

Freelance Geophysicist 2019–2022
Self-employed — Santarém, PA, Brazil

- Processing and interpretation of potential field data for geoscience projects.

Administrative Director 2017–2018
Prospecta Jr. (Junior Enterprise, UFOPA)

- Administrative management and team leadership at the Geophysics junior enterprise.

EDUCATION

Master of Geophysics 2023–2025
National Observatory (Brazil)

Thesis: Interpretation of magnetic data over oceanic crust at low latitudes.

B.Sc. in Geophysics 2017–2019
Federal University of Western Pará (UFOPA)

Final project: Joint inversion of RF and DOS in the southeastern portion of the Amazon Craton.

B.Sc. in Earth Sciences 2013–2017
Federal University of Western Pará (UFOPA)

Final project: Binary segmentation for interpretive enhancement of geophysical well log profiles.

PUBLICATIONS

Submitted Manuscript

- Oliveira Jr., V.C.; **Sousa Jr., R.O.**; Barbosa, V.C.F.; Maia, M. (2026). Regional magnetic inversion of marine trackline data. *Journal of Geophysical Research: Solid Earth* (submitted).

Journal Article

- Sousa Jr., R.O.** ; Alves, A.L. (2020). Lithostratigraphic correlation — Brazos-Trinity, Gulf of Mexico. *Brazilian J. of Development*. DOI: 10.34117/bjdv6n3-054

Conference Proceedings

- Sousa Jr., R.O.**; Trindade, C.R.D.; Soares, J.E. (2018). Crustal structure of the Amazon Craton — joint inversion of RF and DOS. *SBGf*. DOI: 10.22564/8simbgf2018.115

ADDITIONAL COURSES

